

Efficiency of Giant Panda and Snow Leopard as Umbrella Species

Nicholas Chen, Princeton University

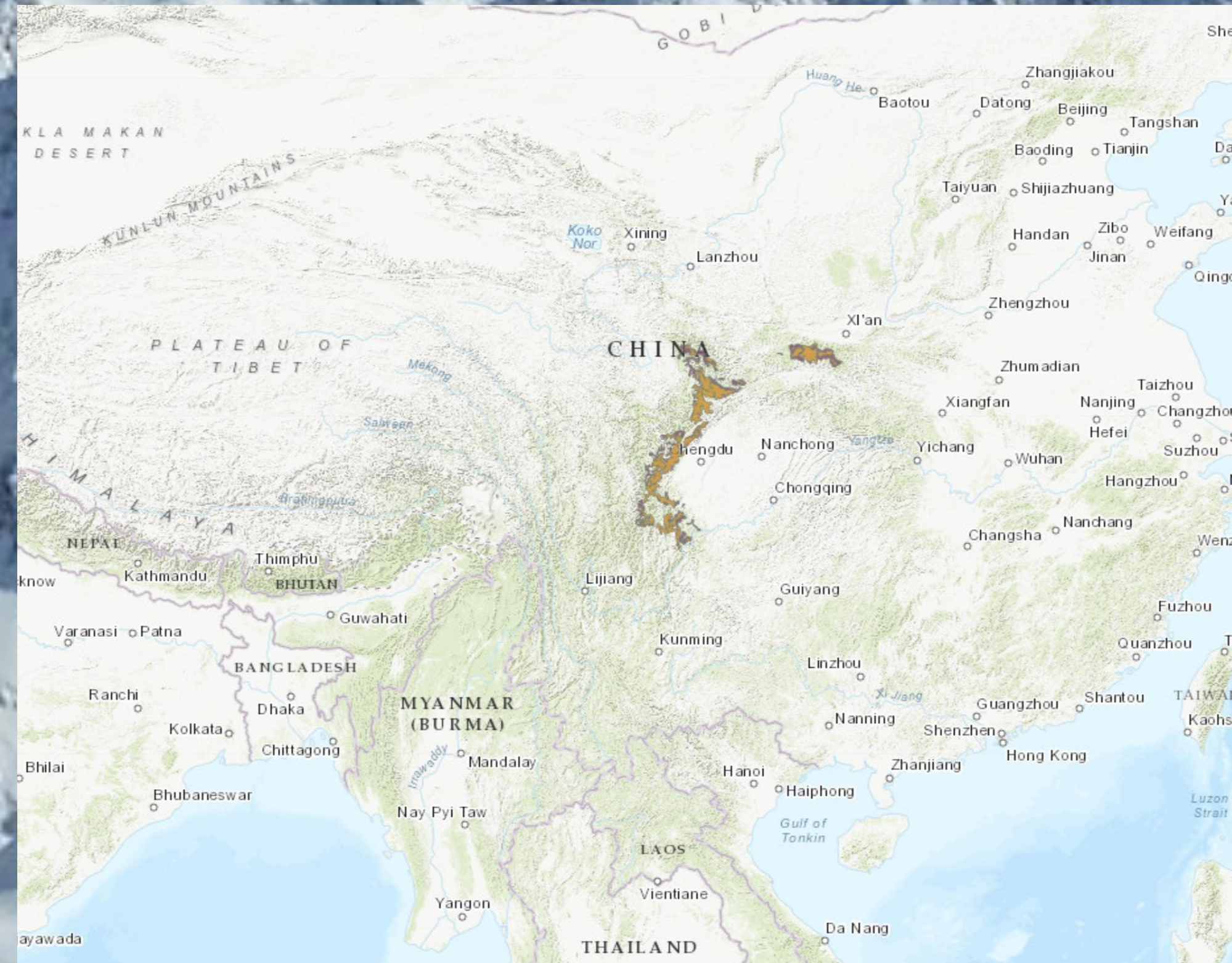
Background and Goals

The giant panda has long served as China's flagship species, raising conservation awareness and funding [1]. However, despite this success, its habitat has failed to protect many large carnivores, such as snow leopards and wolves, leading to significant population declines [2]. This research evaluates whether adding the snow leopard as a second flagship species could improve biodiversity conservation.

The goal is to assess both species' effectiveness as umbrella species—those whose protected habitats benefit many other species—by comparing their species coverage per square kilometer and cost-efficiency [3][4][5].



A picture of a Giant Panda Creator: guenterguni | Credit: Getty Images



Giant Panda Habitat Range Credit: IUCN 2025

Methods and Data

The study used IUCN Red List data [6] to calculate:

- Species per km² protected by each species' habitat
- Cost per km² to maintain each habitat

Only the snow leopard's range within China was considered [8].

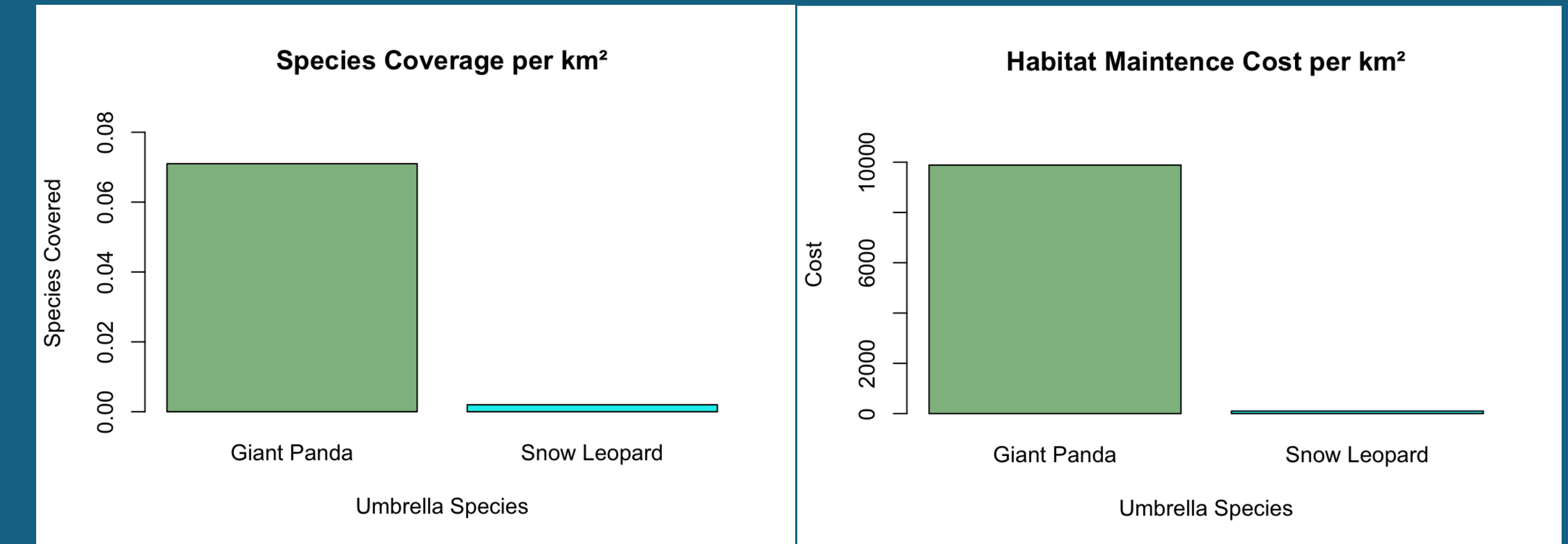
- Cost for Giant Panda: \$255 million/year for giant pandas [7]
- Cost for Snow Leopard \$200 million [8].
- Species coverage focused on critically endangered, endangered, and vulnerable species, and calculations assumed uniform distribution across habitat ranges [9].

Discussion

- Giant panda: Protects more species per km² (0.071), but at a higher cost (\$9,884/km²)
- Snow leopard: Protects fewer species per km² (0.002), but is significantly cheaper (\$100/km²) and overlaps with more critically endangered species [6].

Despite its lower spatial efficiency, the snow leopard's cost-effectiveness and higher coverage of priority species make it a strong candidate for co-flagship status. Previous studies have emphasized the snow leopard's public and ecological appeal [5][6], but this study adds a critical economic and spatial efficiency dimension. The findings support a dual flagship strategy to improve biodiversity outcomes while ensuring sustainable use of conservation resources.

Figure/Results



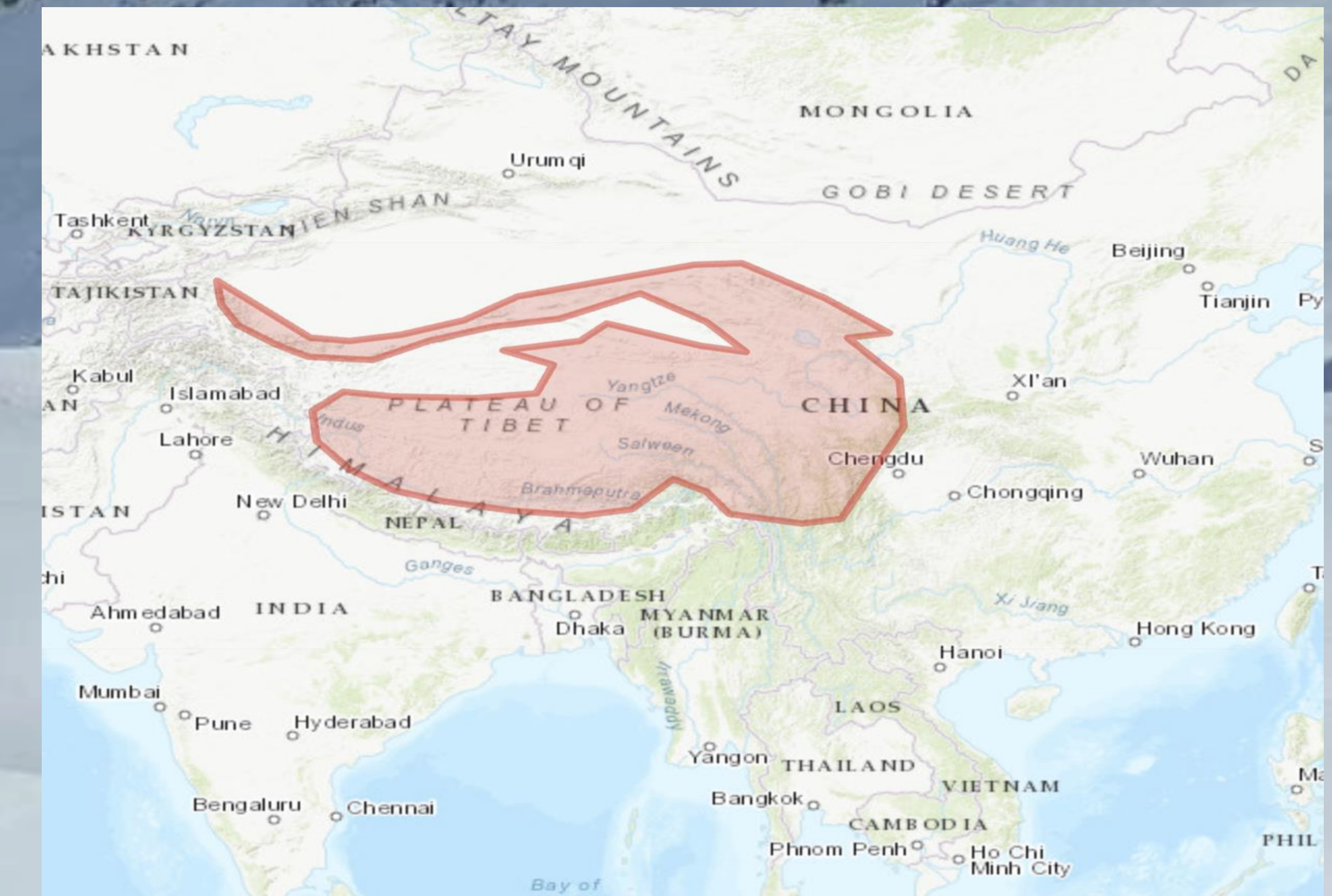
Type Of Species Covered

Coverage by	Critically Endangered	Endangered	Vulnerable
Giant Panda	13	44	69
Snow Leopard	20	53	88
Both	9	28	53

Graph produced with Rstudio Team 2024



A picture of a Snow Leopard Don Johnston | Credit: Getty Images/All Canada Photos



Snow Leopard Habitat Range Credit: IUCN 2025

References

- Wei, F., Costanza, R., Dai, Q., Stoeckl, N., Gu, X., Farber, S., ... Zhang, W. (2018). The value of ecosystem services from giant panda reserves. *Current Biology*, 28(13), 2174–2180.e7. <https://doi.org/10.1016/j.cub.2018.05.046>
- Li, S., McShea, W. J., Wang, D., Gu, X., Zhang, X., Zhang, L., & Shen, X. (2020). Retreat of large carnivores across the giant panda distribution range. *Nature Ecology & Evolution*, 4(10), 1327–1331. <https://doi.org/10.1038/s41559-020-1260-0>
- Simberloff, D. (1998). Flagships, umbrellas, and keystones: Is single-species management passé in the landscape era? *Biological Conservation*, 83(3), 247–257. [https://doi.org/10.1016/S0006-3207\(97\)00081-5](https://doi.org/10.1016/S0006-3207(97)00081-5)
- Yue, Y., Wang, Y., Ye, Z., Zhang, C., Qiu, L., Xu, Q., ... Dai, Q. (2024). Ecological and public advantages of a dual flagship strategy: Giant panda and snow leopard. *Diversity*, 16(2), Article 2. <https://doi.org/10.3390/d16020076>
- Kau, M., Weckworth, B. V., Li, S., Pires, M. M., Jin, D., Pacifici, M., ... Li, J. (2025). Umbrella, keystone, or flagship? An integrated framework for identifying effective surrogate species. *Biological Conservation*, 303, 111025. <https://doi.org/10.1016/j.biocon.2025.111025>
- IUCN. (2025). The IUCN Red List of Threatened Species: Species data download. <https://www.iucnredlist.org>
- Snow Leopard Facts: Diet, Habit, Threats & Conservation. (n.d.). International Fund for Animal Welfare (IFAW). Retrieved April 9, 2025, from <https://www.ifaw.org/animals/snow-leopards>
- Hu, L., Feng, B., Zhang, J., Dong, X., Tang, J., Zhou, C., ... Bai, W. (2024). Impacts of land-use change on the habitat suitability and connectivity of giant panda. *Global Ecology and Conservation*, 53, e03019. <https://doi.org/10.1016/j.gecco.2024.e03019>
- Saving Snow Leopards – Not Just a Day's Work. (n.d.). World Bank. <https://doi.org/10.23/saving-snow-leopards-not-just-a-days-work>
- R Core Team. (2024). R: A language and environment for statistical computing (Version 4.3) [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>
- RStudio Team. (2024). RStudio: Integrated development environment for R (Version 2024.12.1+563) [Computer software]. Posit Software, PBC. <https://posit.co/download/rstudio-desktop/>